Assignment 1: Math1014

September 9, 2026

1 Linear Algebra

1.1 Question 1

we are given the vector space

$$V_N = \operatorname{span}\{\cos(2\pi kt), \sin(2\pi kt) \mid 0 \leq k \leq N\}.$$

(a)

two functions are linearly independent if the only scalars a, b satisfying

$$a\cos(2\pi t) + b\sin(2\pi t) = 0 \quad \text{for all } t \in [0,1]$$

are $a = b = 0$. if we can find even one nontrivial choice of t that forces $a = 0$, and another that forces $b = 0$, we're done.
- at $t = 0$,

$$a\cos(0) + b\sin(0) = a(1) + b(0) = a = 0$$

so a must be 0
- at $t = \frac{1}{4}$

$$a\cos\left(\frac{\pi}{2}\right) + b\sin\left(\frac{\pi}{2}\right) = a(0) + b(1) = b = 0$$

so b must be 0 too
since the equation forces $a = b = 0$, the only linear combination equal to the zero function is the trivial one. hence $\cos(2\pi t)$ and $\sin(2\pi t)$ are **linearly independent** in V_1.

(b)

we need the dimension of V_1
the definition says

$$V_N = \operatorname{span}\{\cos(2\pi kt), \sin(2\pi kt) \mid 0 \leq k \leq N\}$$

if $N = 1$, then k can be either 0 or 1.
so we get

$$V_1 = \text{span}\{\cos(0), \sin(0), \cos(2\pi t), \sin(2\pi t)\}$$

now, $\cos(0) = 1$ and $\sin(0) = 0$
so the span becomes:

$$V_1 = \text{span}\{1, \cos(2\pi t), \sin(2\pi t)\}$$

the zero function doesn't add anything to the span, so we can ignore it
we therefore have three functions:

$$1, \cos(2\pi t), \sin(2\pi t)$$

these are linearly independent, so they form a basis for V_1.
therefore, $\boxed{dim(V_1) = 3}$

(c)

finding the dimension of V_2,
when $N = 2$, the possible values of k are $0, 1, 2$
therefore,

$$V_2 = \text{span}\{\cos(0), \sin(0), cos(2\pi t), sin(2\pi t), cos(4\pi t), sin(4\pi t)\}$$

again, $\cos(0) = 1$ and $\sin(0) = 0$, so (leaving out the 0 terms),

$$V_2 = \text{span}\{1, cos(2\pi t), sin(2\pi t), cos(4\pi t), sin(4\pi t)\}$$

there are now 5 independent functions. therefore,

$$\boxed{dim(V_2) = 5}$$

for a general N, we always have:

- one constant function 1,
- N cosine functions corresponding to $k = 1, ..., N$,
- n sine functions coreresponding to $K = 1, ..., N$

so, the total number of basis functions is

$$1 + N + N$$

therefore,

$$\boxed{dim(V_N) = 2N + 1}$$

1.2 Question 2

we have $S : V_N \to R_m$ defined by

$$S(f) = \begin{bmatrix} f(t_1) \\ f(t_2) \\ \vdots \\ f(t_m) \end{bmatrix}$$

(a)

we are given $m = 3$ with sampling points $t_1 = 0$, $t_2 = 0.5$, $t_3 = 1$ and

$$f(t) = 3 + \cos(2\pi t)$$

since S records the value of f at each sampling point,

$$S(f) = \begin{bmatrix} f(0) \\ f(0.5) \\ f(1) \end{bmatrix}$$

- at $t = 0$,

  $$f(0) = 3 + \cos(2\pi(0))$$
  $$f(0) = 3 + \cos(0)$$

  since $\cos(0) = 1$,

  $$f(0) = 4$$

- at $t = 0.5$,

  $$f(0.5) = 3 + \cos(2\pi(0.5))$$
  $$f(0.5) = 3 + \cos(\pi)$$

  since $\cos(\pi) = -1$,

  $$f(0.5) = 3 - 1 = 2$$

- at $t = 1$,

  $$f(1) = 3 + \cos(2\pi(1))$$
  $$f(1) = 3 + \cos(2\pi)$$

  since $\cos(2\pi) = 1$,

  $$f(1) = 3 = 1 = 4$$

putting all these together into the matrix, we get

$$\boxed{S(f) = \begin{bmatrix} 4 \\ 2 \\ 4 \end{bmatrix}}$$

(b)

now we have

$$g(t) = \sin(12\pi t)$$

again,

$$S(g) = \begin{bmatrix} g(0) \\ g(0.5) \\ g(1) \end{bmatrix}$$

- at $t = 0$,
$$g(0) = \sin(12\pi(0)) = \sin(0) = 0$$
- at $t = 05$,
$$g(0.5) = \sin(12\pi(0.5)) = \sin(6\pi) = 0$$
- at $t = 1$,
$$g(1) = \sin(12\pi(1)) = \sin(12\pi) = 0$$

putting all these together into the matrix, we get

$$\boxed{S(f) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}$$

(c)

for a function to be a linear map, we need

$S(af + bg) = aS(f) + bS(g)$ for any functions $f, g \in V_N$ and scalars $a, b \in \mathbb{R}$

starting with the LHS,

$$S(f) = \begin{bmatrix} (af+bg)(t_1) \\ (af+bg)(t_2) \\ \vdots \\ (af+bg)(t_m) \end{bmatrix}$$

we know that at each sample point t_i, the addition and scalar multiplication is

$$(af + bg)(t_i) = af(t_i) + bg(t_i)$$

therefore,

$$S(f) = \begin{bmatrix} af(t_1) + bg(t_1) \\ af(t_2) + bg(t_2) \\ \vdots \\ af(t_m) + bg(t_m) \end{bmatrix}$$

separating into two vectors:

$$S(f) = a \begin{bmatrix} f(t_1) \\ f(t_2) \\ \vdots \\ f(t_m) \end{bmatrix} + b \begin{bmatrix} g(t_1) \\ g(t_2) \\ \vdots \\ g(t_m) \end{bmatrix}$$

we can see that these two vectors are exactly $S(f)$ and $S(g)$
therefore,

$$S(af + bg) = aS(f) + bS(g)$$

hence,

$$\boxed{\text{S is linear}}$$

(d)

an element of V_N can be represented as a vector in $\mathbb{R}^{2N+1}$
from question 1, $dim(V_N) = 2N + 1$, so $[f]_{\mathcal{B}} \in \mathbb{R}^{2N+1}$.
so there are $2N+1$ functions in this basis, meaning $[f]_{\mathcal{B}}$ has $2N+1$ components
S takes f and evaluates it at m sample points

$$S(f) = \begin{bmatrix} f(t_1) \\ f(t_2) \\ \vdots \\ f(t_m) \end{bmatrix}$$

therefore, $S(f) \in \mathbb{R}^m$. so the output has m components
so we need a matrix A such that

$$\text{matrix } A \times \text{input vector} = \text{output vector}$$

therefore, A needs to take a vector with $2N+1$ entries and turn it into a vector with m entries.
this means A needs $2N+1$ columns (one for each basis function) and m rows (one for each sample point).
therefore,

$$\boxed{A \in \mathbb{R}^{m \times (2N+1)}}$$

(e)

we know from 2(d) that

$$A \in \mathbb{R}^{m \times (2N+1)}$$

so A has m rows and $2N+1$ columns.
the rank of a matrix cannot be bigger than the number of rows or the number of columns.

therefore,

$$rank(A) \leq m$$

and

$$rank(A) \leq 2N + 1$$

so the upper bound is the smaller of these two numbers:

$$\boxed{rank(A) \leq \min(m, 2N + 1)}$$

this is because the rank tells us how many independent rows or columns there are, so we cannot have more independent rows than m, or more independent columns than $2N + 1$.

(f)

we can use the rank-nullity theorem:

$$rank(A) + nullity(A) = \text{number of columns of } A$$

from 2(d), A has $2N + 1$ columns, so

$$rank(A) + nullity(A) = 2N + 1$$

rearranging,

$$nullity(A) = 2N + 1 - rank(A)$$

from part (e),

$$rank(A) \leq \min(m, 2N + 1)$$

so the smallest possible nullity occurs when the rank is as large as possible. therefore,

$$nullity(A) \geq 2N + 1 - \min(m, 2N + 1)$$

so,

$$\boxed{nullity(A) \geq \max(0, 2N + 1 - m)}$$

if $m < 2N + 1$, then

$$\boxed{nullity(A) \geq 2N + 1 - m > 0}$$

(g)

from part(f) we know that

$$nullity(A) \geq 2N + 1 - m.$$

we want

$$dim(null(A)) > 0.$$

this will definitely happen when

$$2N + 1 - m > 0.$$

so,

$$2N + 1 > m.$$

therefore,

$$\boxed{m < 2N + 1}$$

leads to

$$\boxed{dim(null(A)) > 0}.$$

in other words, if we have fewer sample points than the number of basis functions, there must be some non-zero functions that get mapped to the zero vector.

(h)

if

$$dim(null(A)) > 0$$

then the null space of A contains some non-zero vector.
this means there is some $f \neq \vec{0}_{V_N}$ such that

$$A[f]\mathcal{B} = \vec{0}\mathbb{R}^m$$

but we know that

$$A[f]_{\mathcal{B}} = S(f)$$

so this means

$$S(f) = \vec{0}$$

therefore, a non-zero signal f can be sampled and produce exactly the same samples as the zero signal. this means that S is not one-to-one, because different signals can give the same sampled output.
for example, if

$$S(f) = \vec{0}$$

then for any signal g,

$$S(g + f) = S(g) + S(f)$$

because S is linear.

therefore,

$$S(g + f) = S(g)$$

so we cannot distinguish between g and $g + f$ from the sampled data.
this means there is a loss of information, which is related to the idea of aliasing.
therefore S is not injective and the samples do not uniquely determine the original signal.

2 Calculus

2.1 Question 3

(a)

we have

$$f(x) = \cos(x) + \sin(x)$$

to find the 4th-degree Taylor polynomial centred at $a = 0$, we use

$$P_4(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4$$

first, find the derivatives:

$$f(x) = \cos(x) + \sin(x)$$

$$f'(x) = -\sin(x) + \cos(x)$$
$$f''(x) = -\cos(x) - \sin(x)$$
$$f'''(x) = \sin(x) - \cos(x)$$
$$f^{(4)}(x) = \cos(x) + \sin(x)$$

now evaluate each one at $x = 0$:

$$f(0) = 1$$

$$f'(0) = 1$$
$$f''(0) = -1$$
$$f'''(0) = -1$$
$$f^{(4)}(0) = 1$$

substituting these into the Taylor formula gives

$$P_4(x) = 1 + x - \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!}$$

therefore,

$$\boxed{P_4(x) = 1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24}}$$

(b)

we are sampling the signal at $t_1 = 0.1$
so we use $P_4(0.1)$ to estimate the actual value

$$1 + (0.1) - \frac{(0.1)^2}{2} - \frac{(0.1)^3}{6} + \frac{(0.1)^4}{24}$$

calculating each term gives

$$1 + 0.1 - 0.005 - 0.0001667 + 0.00000417$$

therefore,

$$\boxed{P_4(0.1) \approx 1.09484}$$

so the estimated sample value is approximately

$$\boxed{f(0.1) \approx 1.09484}$$

for the remainder, Taylor's formula gives

$$R_4(x) = \frac{f^{(5)}(c)}{5!}x^5$$

for some c between 0 and x.
we already know

$$f^{(4)}(x) = \cos(x) + \sin(x)$$

so

$$f^{(5)}(x) = -\sin(x) + \cos(x)$$

therefore,

$$\frac{-\sin(c) + \cos(c)}{5!}x^5$$

where $0 \leq c \leq x \leq 0.1$
so,

$$\boxed{R_4(x) = \frac{-\sin(c) + \cos(c)}{120}x^5, \qquad 0 \leq c \leq x \leq 0.1}$$

(c)

we have

$$g(x) = e^{-x}f(x)$$

and

$$f(x) = \cos(x) + \sin(x)$$

therefore,

$$g(x) = e^{-x}(\cos(x) + \sin(x))$$

so the integral is

$$I = \int_0^\infty e^{-x}(\cos(x) + \sin(x)), dx$$

notice that

$$e^{-x}\cos(x) + e^{-x}\sin(x).$$

therefore,

$$I = \int_0^\infty e^{-x}(\cos(x) + \sin(x)), dx$$

can be written as

$$I = \left[-e^{-x}\cos(x)\right]_0^\infty$$

as $x \to \infty$,

$$e^{-x} \to 0$$

while $\cos(x)$ stays between -1 and 1, so

$$-e^{-x}\cos(x) \to 0$$

at $x = 0$, $-e^0 \cos(0) = -1$
therefore,

$$I = 0 - (-1) = 1$$

so the improper integral converges and

$$\boxed{I = 1}$$

2.2 Question 4

(a)

we are given

$$y(t) = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)!}$$

this matches the Taylor series for $\sin(t)$:

$$t - \frac{t^3}{3!} + \frac{t^5}{5!} - \frac{t^7}{7!} + \cdots$$

so the elementary function represented by the series is

$$\boxed{y(t) = \sin(t)}$$

the Taylor series for $\sin(t)$ converges for every real value of t
therefore, its radius of convergence is

$$\boxed{R = \infty}$$

(b)

for a parametric curve, the arc length is

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}, dt$$

we have

$$x(t) = t$$

so

$$\frac{dx}{dt} = 1$$

from **part (a)**,

$$y(t) = \sin(t)$$

so

$$\frac{dy}{dt} = \cos(t)$$

the interval is

$$0 \leq t \leq \pi$$

therefore,

$$\int_0^\pi \sqrt{1^2 + \cos^2(t)}, dt$$

so,

$$\boxed{L = \int_0^\pi \sqrt{1 + \cos^2(t)}, dt}$$

(c)

we only want to estimate the arc length over $0 \leq t \leq 0.5$
from the question, we have

$$\cos(t) \approx 1 - \frac{t^2}{2}$$

therefore,

$$\cos^2(t) \approx \left(1 - \frac{t^2}{2}\right)^2$$

expanding this gives

$$\cos^2(t) \approx 1 - t^2 + \frac{t^4}{4}$$

so,

$$1 + \cos^2(t) \approx 2 - t^2 + \frac{t^4}{4}$$

factor out 2:

$$1 + \cos^2(t) \approx 2\left(1 - \frac{t^2}{2} + \frac{t^4}{8}\right)$$

therefore,

$$\sqrt{1+\cos^2(t)} \approx \sqrt{2}\sqrt{1-\frac{t^2}{2}+\frac{t^4}{8}}$$

using

$$\sqrt{1+u} \approx 1+\frac{u}{2}$$

we get

$$\sqrt{1+\cos^2(t)} \approx \sqrt{2}\left(1-\frac{t^2}{4}+\frac{t^4}{16}\right)$$

therefore, the arc length is approximately

$$L \approx \sqrt{2}\int_0^{0.5}\left(1-\frac{t^2}{4}+\frac{t^4}{16}\right), dt$$

integrating each term gives

$$L \approx \sqrt{2}\left[t-\frac{t^3}{12}+\frac{t^5}{80}\right]_0^{0.5}$$

substituting $t = 0.5$:

$$L \approx \sqrt{2}\left(0.5-\frac{(0.5)^3}{12}+\frac{(0.5)^5}{80}\right)$$

calculating this gives

$$\boxed{L \approx 0.69294}$$